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328352(28)

B. E. (Third Semester)
EXAMINATION, April-May, 2015
(New Scheme)
(Branch : Et & T)

PROBABILITY AND RANDOM VARIABLES

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any *two* from (b), (c), and (d).

1. (a) State the Parseval's theorem. 2

(b) Prove that autocorrelation of periodic signal is : 7

$$R(\tau) = V_0^2 + 2\sum_{n=1}^{\infty} |V_n|^2 \cos\left(\frac{2\pi n\tau}{T}\right)$$

where T is time period of signal.

(c) Find the exponential Fourier series coefficient and power spectral density of square wave shown in Fig. 7

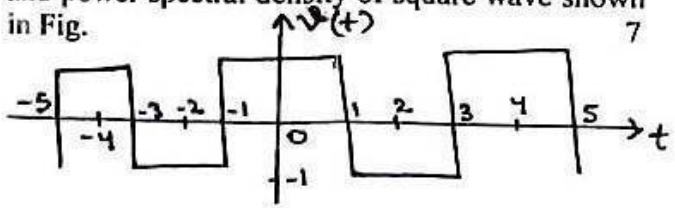

Fig. 1

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(d) Find the Fourier transform and plot magnitude spectrum and phase spectrum of signum function which is defined as :

$$\text{sign}(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ -1 & \text{for } t < 0 \end{cases}$$

2. (a) Define Null and Universel set.

(b) An experiment consists of rolling a signle die. two events are defined as A = {a 6 shows up} and B {a 2 or a 6 shows up} :

(i) Find P (A) and P (B).

(ii) Define a third event (So that P (C) – 1 – P (A) – P (B).

(c) In communication system the signal sent from point 'a' to point 'b' arrives by two path in parallel. Over each path the signal passes through two epeaters (in series). Each repeater in one path has a probability of failing (becoming an open circuit) of 0.005. This probability is 0.008 for each repeater on the other path. All repeaters fail independently of each other. Find the probability that the signal will not arrive at point 'b'. 7

(d) A rifleman can achieve a "marksman" award if he passes a test. He is allowed to fire six shot at a target's bull's eye. If he hits bull's eye with at least five of his six shots he wins a set. He becomes a markman only if he can repeat the feat three straight, that is if he can win three straight

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sets. If his probability is 0.8 of hitting a bull's eye on any *one* set, find the probabilities of (i) winning a set (ii) becoming a marksman. 7

3. (a) Give the relation between probability distribution and probability density function. 2

(b) Consider the probability density function $f_x(x) = a\,e^{-b|x|}$ where x is random variable and range from $-\infty + 0 + \infty$. 7

(i) Find the CDF $F_x(x)$.

(ii) The relationship between 'a' and 'b'.

(iii) The probability that the outcome x lies between 1 and 2.

(c) Explain the Reyleigh and Gaussian probability density function 7

(d) If Z = X + Y where X and Y are independent random variable then find mean and variance of random variable 'Z' in term of mean and Variance of Random variable X and Y. 7

4. (a) When is random process called ergodic in mean random process ? 2

(b) If $R_{XY}(\tau)$ is autocorrelation of wide sense stationary Random process X (t), then prove that : 7

(i) $R_{XX}(0)$ is mean square value of Random process.

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(ii) $R_{XX}(\tau)$ is even function.

(iii) $|R_{XX}(\tau)| \le R_{XX}(0)$.

(c) Telephone calls are initiated through an exchange at the average rate of 75 per minute and are described by a Poisson process. Find the probability that more than 3 calls are initiated in any *five* second period. 7

(d) Let $X(t)$ be a wide sense stationary process with autocorrelation function. 7

$R_{XX}(\tau) = e^{-a|\tau|}$ where $a > 0$ is constant and assume $X(t)$ 'amplitude modulates' a carrier $\cos(\omega_0 t + \theta)$ as shown in Fig. 2. Where ω_0 is a constant and θ is a random variable uniform on $(-\pi, \pi)$ that is statistically independent of X (t). Determine and plot autocorrelation function of Y (t).

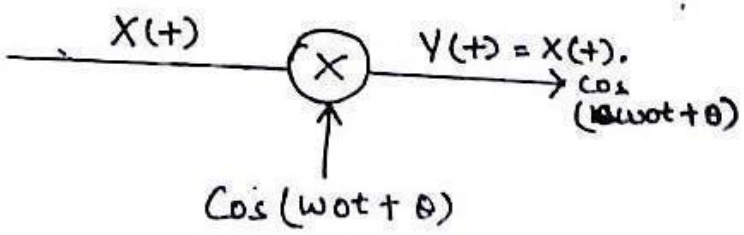

Fig. 2

5. (a) Write the relationship between cross-power spectrum and cross-correlation function. 2

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(b) Define cross-power density and what are the properties of cross-power density. 7

(c) Derive the relationship between power spectrum and autocorrelation function. 7

(d) Consider the random process $X(t) = A_0 \cos(\omega_0 t + \theta)$ where A_0 and ω_0 are real constant and θ is a random variable uniformally distributed on the interval $(0, \pi/2)$. Find the average power P_{xx} in $X(t)$. 7

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